THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Dilute solution has: 1

(a) less solute **(b)** no solvent **(c)** no solute **(d)** more solute

2. Solvent is the substance that: 1

(a) gets dissolved **(b)** is a gas only
(c) settles **(d)** dissolves the solute

3. Separation of oil and water by: 1

(a) sublimation **(b)** separating funnel
(c) magnet **(d)** filtration

4. Concentrated solution has: 1

(a) no solute **(b)** no solvent **(c)** more solute **(d)** less solute

5. Water is the solvent in: 1

(a) alloy **(b)** nail polish
(c) tincture of iodine **(d)** salt solution

6. Husk from grain by: 1

(a) winnowing **(b)** distillation **(c)** filtration **(d)** sublimation

7. Air can be liquefied by: 1

(a) heating **(b)** filtration
(c) sublimation **(d)** pressure and cooling

8. Tyndall effect is shown by: 1

(a) true solutions **(b)** air in a room **(c)** colloids **(d)** pure water

9. Crude oil is separated by: 1

(a) filtration **(b)** sublimation
(c) fractional distillation **(d)** winnowing

10. Chromatography means: 1

(a) water writing **(b)** colour writing **(c)** air writing **(d)** heat writing

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the Tyndall effect. 2

12. What is a colloid? 2

13. What is distillation? 2

14. Name two alloys. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the steps to separate salt and sand. 3

16. Write three applications of chromatography. 3

17. Why do we use a condenser in distillation? Give three points. 3

OR

Find the percentage of solute: 15 g sugar in 85 g water. 3

18. Why is water called a universal solvent? Give three points. 3

OR

Name the items used in the paper chromatography experiment. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain crystallisation. 5

OR

Write the concentrations of solutions with formulas. 5

20. Describe the properties of colloids. 5

OR

Explain alloys with uses. 5

END OF QUESTION PAPER — Total: 40 Marks

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